

Qualia Theorems

1. The math of subjectivity

If we use category theory as Jon Zingale proposed, then “intersubjectivity is a colimit of subjectivity and objectivity is a limit of interobjectivity” as Jon puts it. In plain English this means intersubjectivity is achieved by adding distinct personal voices together (the colimit), while objectivity and objective truth is distilled by filtering out personal variation (the limit).

If we use topology and Cobordism, subjective individual perspectives can be modeled as lower-dimensional manifold boundaries (e.g., (n-1)-dimensional spaces S_1 and S_2 , while the shared intersubjective space is the n-dimensional cobordism manifold W connecting them.

If we use gauge theory, we are treating objectivity not as an absolute “view from nowhere” but as the invariant structure underlying all local perspectives. Subjectivity can be described as symmetry breaking and objectivity as gauge invariance. A perspective switch for 1st-person views from person A to person B becomes a gauge transformation. For local gauge invariance to hold across different points in space, a field (like the photon field in QED) must exist to connect them. Tools or places that bring people closer together and align their views act as a similar kind of a gauge field (or “reality distortion field”). Cinemas for example are places where world views converge. All members of the audience share for a while the same world view, they view and hear the same story which influences their world view (what is good, what is bad, what should be avoided, etc.). If the story is convincing the audiences adopts part of it and adjust the own world view.

Viewpoint	Perspective Switch	Subjectivity → Intersubjectivity	Interobjectivity → Objectivity
Category Theory	Natural Isomorphism / Automorphism	intersubjectivity is a colimit of subjectivity. Colimit: distinct subjective objects are glued together along shared boundaries	objectivity is a limit of interobjectivity Limit: Filtering down to the invariant core that holds across all perspectives/ relationships
Topology (Cobordism)	A Homeomorphism acting along the boundary manifolds ∂W	Boundary to Interior: $S_1 \sqcup S_2 = \partial W$. Individual minds (S_1, S_2) are glued into a shared manifold (W).	Topological Invariants: The intrinsic geometric properties of W that remain unchanged regardless of perspective
Gauge Theory	Gauge Transformation $U(x)$	intersubjectivity is a gauge field Gauging a Local Symmetry & Introducing the Connection Field (A_μ) $D_\mu = \partial_\mu - igA_\mu$	objectivity is a gauge-invariance Guage Orbit Quotient Space ($\mathcal{M}/G$) is the geometric manifold remaining after factoring out all local gauge degrees of freedom

The version of the hard problem that matters pragmatically is translating between two different 1st-person perspectives. If we can solve this problem, we can use it to solve the more general problem: each of us knows what it is like to be a human that experiences pleasure or pain, hunger or disgust, love or fear. These are the basic irreducible dimensions of subjective experience. If we learn what combination or mix the other person has experienced how and why and translate that 1st-person experience from the other person to ourselves, we begin to understand what-it-is-like-to-be that person.

This problem has no exact solution, only a quantified approximate one. Exact equivalence of subjective experience is generically unattainable and almost never happens; but sharing a path gives us a approximation down to a floor that is almost surely above zero.

Scale / Domain	Philosophical Concept	Gauge Theory Model	Mathematical / Physical Counterpart
1st-Person	Subjectivity	Local Gauge Fixing	Selecting a specific gauge condition (e.g., Lorenz gauge, Coulomb gauge) at x^μ .
1st A $\rightarrow$ 1st B	Perspective Switch	Gauge Transformation	Applying a phase rotation matrix $U(x)$ between two frame choices.
2nd-Person	Intersubjectivity	Covariant Coupling & Connection (A_μ)	Introducing the vector potential A_μ to preserve covariance under local transformations $D_\mu = \partial_\mu - igA_\mu$.
Systemic	Interobjectivity	Networked Gauge Field Dynamics	The full web of field interactions and field-strength tensors ($F_{\mu\nu}$) across all nodes.
3rd-Person	Objectivity	Gauge Orbit Quotient Space ($\mathcal{M}/\mathcal{G}$)	Moduli space geometry: the invariant manifold obtained by quotienting out all gauge redundancies.

Using a gauge-theoretic formulation borrowed from physics (where it is popular in high-energy physics and Quantum Field Theories) we can formulate these two theorems.

LNC Impossibility Theorem: exact equivalence almost never happens

Claim. Two independent agents almost never have exactly equivalent worldviews.

Idea. Two worldviews are equivalent if one can be turned into the other by relabeling the emotional axes, which in the model means applying a rotation Q . Picture all possible pairs of worldviews as a big space. The equivalent pairs form only a thin sheet inside it, like a sheet of paper in a room. The sheet has fewer free dimensions than the room, so a randomly chosen pair almost surely misses it.

We use a specific gauge-theoretic formulation, where the emotional valuation matrix (or valuation structure) is modeled as the gauge connection $U: E \rightarrow G$ and its associated holonomy matrices $H_k(U) \in G$. In this mathematical model, the holonomy matrices model a loop (or fundamental cycle) that represents a closed chain of experiences or reasoning that returns to the same starting concept. The holonomy matrix H_k measures the net orientation change of the agent's internal valuation axes after completing that full loop.

The Connection / Valuation Matrix ($U_e \in G$): Each edge e in the context graph represents a specific cognitive or emotional transition (e.g., encountering a situation, stimulus, or memory). The matrix assigned to that edge, U_e , acts as an operator in a compact Lie group G (such as $SO(3)$). In human terms, it describes how an individual's internal state rotates or shifts its "valuation axes" when moving along that experience

The Observables / Holonomies ($H_k \in G$): When an agent undergoes a closed loop of experience (a cycle in the context graph, such as reflecting on a recurring life event or concept), the product of the edge matrices along that cycle forms a holonomy matrix $H_k(U) \in G$. These holonomies represent the agent's net perspective shift across fundamental value loops

What Makes it "Unique" to an Individual: An individual's unique worldview is the total set of these holonomy matrices ($H_1, H_2, \dots, H_r$). Two people have the *same* emotional valuation structure if and only if there exists a global change-of-basis matrix $Q \in G$ that aligns all their loop matrices simultaneously ($H_k(Y) = QH_k(X)Q^{-1}$ for all k).

The entire point of the LNC Impossibility Theorem is proving that because these emotional matrices $H_k(U)$ inhabit a continuous Lie group like $SO(3)$, no two independent human minds will ever possess the exact same set of valuation matrices.

Counting dimensions is the key for the proof: One worldview is modeled as r fundamental emotional/cognitive loops (holonomies). Since $SO(3)$, the symmetry group of 3D rotations, has 3 parameters like pitch, roll, yaw, one worldview takes $3r$ numbers to specify. Two independent worldviews X and Y together require $3r + 3r = 6r$ free numbers. This $6r$ -dimensional space of all pairs is our “room”. X and Y are equivalent only if there is a single perspective transformation Q (3 parameters) that lines up all r loops of X with Y simultaneously. An equivalent pair is built by choosing one worldview (r holonomies, 3 free numbers each in $SO(3)$) and one rotation Q (3 numbers), giving $3r + 3$ free numbers. For $r \geq 2$, $3r + 3 < 6r$, so the sheet is thinner than the room. We could say the worldview is an own ‘slice of the world’. Two randomly chosen sheets or slices almost surely are not identical. For $r = 1$ the counts tie, but rotations around a holonomy’s own axis leave it unchanged, which makes the sheet effectively one dimension thinner.

Asymptotic Qualia Theorem: asymptotic approximation is possible

Claim. If two agents share a path, the gap converges to a floor, and the floor is almost surely above zero. Sharing a path gives asymptotic approximation down to a floor

Idea, part 1. If both agents’ transitions settle down exponentially fast, their holonomies settle down too, and the distance between them settles to the distance between the final states. Each holonomy is a product of at most L edge matrices, so an error on each edge becomes an error of at most L times that on the holonomy. The error dies out exponentially.

Idea, part 2. On loops built only from shared transitions, the two agents end up identical. On the other loops they differ randomly. Equivalence would then need a rotation Q that leaves the shared part unchanged and also lines up the random parts. That is a thin sheet again, so the floor is almost surely above zero.

2. Assumptions

Label	Assumption
G	A compact connected Lie group with $\dim G \geq 1$ (for example $SO(3)$), acting by a faithful unitary representation, so that the Frobenius norm is invariant under multiplication by group elements.
Graph	A finite connected context graph with $r = E - V + 1$ independent cycles. We require $r \geq 1$: if the graph has no cycles, every connection is gauge-trivial and every pair of agents is equivalent.
Worldview	A connection $U: E \rightarrow G$, with U on a reversed edge equal to the inverse. Values are exact (infinite precision).
A1 (density)	The holonomies of the two agents have a joint density with respect to Haar measure (independent, continuously distributed). This is an assumption about the world, not a derived fact.
Bridge	Equivalent experience corresponds to gauge equivalence of valuation structure. This is assumed, not derived.

3. Classification and Distance

Fix a spanning tree and let $H_k(U) \in G, k = 1, \dots, r$, be the holonomies of the fundamental cycles (tree path plus one non-tree edge) at a base vertex.

Fact 1 (standard). Two connections X and Y are gauge equivalent if and only if there is $Q \in G$ with $H_k(Y) = Q H_k(X) Q^{-1}$ for all k .

Define the distance:

$$d(X, Y) = \min_{Q \in G} \max_k \|H_k(Y) - Q H_k(X) Q^{-1}\|$$

- The minimum is attained because G is compact.
- $d = 0$ if and only if X and Y are gauge equivalent.
- d satisfies the triangle inequality (compose the rotations and use the invariance of the norm), so it is a metric on the quotient of connections by gauge.

4. LNC Impossibility Theorem

The LNC Impossibility Theorem says

Assume $r \geq 1$ and A1. Then $P(X \cong Y) = 0$, and therefore $d(X, Y) > 0$ almost surely.

Proof

Let $\Phi: G \times G^r \rightarrow G^r \times G^r$ be $\Phi(Q, h) = (h, QhQ^{-1})$. The set E of equivalent holonomy pairs is the image of Φ . It is compact (a continuous image of a compact set), hence measurable. It suffices to show E has Haar measure zero, since a joint density assigns probability zero to such sets.

Step 1: The kernel of the differential. At a point (Q, h) , the differential of Φ vanishes on a direction $(\xi, 0)$ exactly when ξ commutes with every h_i , that is, ξ lies in the Lie algebra of the centralizer of h . Writing $s(h)$ for its dimension, the rank of $d\Phi$ is $(r + 1) \cdot \dim G - s(h)$.

Step 2: Case $r \geq 2$. The domain has dimension $(r + 1) \cdot \dim G$, which is smaller than the target dimension $2r \cdot \dim G$. The rank is below the target dimension at every point, so every point is critical. By Sard's theorem, the image has measure zero.

Step 3: Case $r = 1$. Domain and target have the same dimension $2 \cdot \dim G$. Every element h of a compact connected group lies in a maximal torus, whose Lie algebra commutes with h and has dimension $\text{rank}(G) \geq 1$. Hence $s(h) \geq 1$, the rank is at most $2 \cdot \dim G - 1$ everywhere, every point is critical, and Sard's theorem again gives measure zero.

Step 4: $d > 0$. Since the minimum in d is attained, $d = 0$ would mean exact equivalence, which has probability zero. ■

Numerical check of the dimension counts

Check of Dimension of $E = (r + 1) \cdot \dim G - s(h) = (r + 1) \cdot 3 - s(h)$ where $G = \text{SO}(3)$, so $\dim G = 3$

$s(h)$ is the set of commuting directions. It is defined as the dimension of the Lie algebra of the joint centralizer (or stabilizer) of the holonomies $(h_1, \dots, h_r)$ inside G

$$s(h) = \dim\{\xi \in \mathfrak{g} \mid \xi h_i = h_i \xi \text{ for all } i = 1, \dots, r\}$$

r	Domain dimension 3r+3	Target dimension 6r	Dimension of E (Jacobian rank)
1	6	6	5
2	9	12	9
3	12	18	12
4	15	24	15

The numbers match the formula $(r + 1) \cdot \dim G - s$ with $s(h) = 1, 0, 0, 0$ respectively. For a finite group G , or without **A1**, the statement fails, and the probability of equivalence is positive.

If we model internal emotional or cognitive states using an N-dimensional Lie group like $\text{SO}(N)$ (where $N = 10, 100$, or 1000), then the dimension of the rotation group $\text{SO}(N)$ is given by the formula:

$$\dim \text{SO}(N) = \frac{N(N-1)}{2}$$

As human cognitive or AI valuation spaces grow into hundreds or thousands of dimensions, the space of possible worldviews expands so rapidly that the slice of “equivalent worldviews” becomes infinitesimally thinner.

Dimensions N	Lie Group G	$\dim G = \dim \text{SO}(N)$	Domain dimension 3 $\dim G$	Target dimension 4 $\dim G$
3	SO (3)	3	9	12
10	SO (10)	45	135	180
100	SO (100)	4,950	14,850	19,800
1,000	SO (1,000)	499,500	1,498,500	1,998,000

5. Asymptotic Qualia Theorem

B1: convergence to the distance of the limits

Suppose each agent's edge matrices satisfy $\|U_e(t) - U_e(\infty)\| \leq \delta_0 e^{-\gamma t}$ for all edges e .
 Let L be the maximum length of a fundamental cycle and let $\epsilon_0 = d(X_\infty, Y_\infty)$.
 Then $|d(X_t, Y_t) - \epsilon_0| \leq 2L \delta_0 e^{-\gamma t}$

Proof.

For unitary matrices, $\|A_1 \cdots A_L - B_1 \cdots B_L\| \leq \sum \|A_i - B_i\|$ (telescoping, using the invariance of the norm). So each holonomy moves by at most $L \delta_0 e^{-\gamma t}$.

For any fixed Q , the quantity $\max_k \|H_k(Y) - QH_k(X)Q^{-1}\|$ changes by at most the sum of the two holonomy perturbations, because conjugation by Q is an isometry. Taking the minimum over Q preserves the bound. ■

What B1 does and does not say. It is continuity. The exponential convergence of the transitions is assumed; it is not derived from a model of shared experience. It also does not show that the distance decreases monotonically or that $d(X_0, Y_0) > \epsilon_0$.

B2: the floor is almost surely positive

Setup. Split the fundamental cycles into I_s (cycles built only from shared transitions, whose limit holonomies are identical for both agents) and I_u (the remaining cycles, with $r_u = |I_u| \geq 1$). Assume that conditional on the shared holonomies h_s , the pairs $(H_k(X_\infty), H_k(Y_\infty))$ for k in I_u have a joint density with respect to Haar measure.

Then $\epsilon_0 = d(X_\infty, Y_\infty) > 0$ almost surely, so $d(X_t, Y_t)$ converges to a strictly positive floor.

Proof. Let K be the centralizer of h_s in G , that is, the rotations Q with $QH_kQ^{-1} = H_k$ for all $k \in I_s$. K is a compact subgroup. Equivalence of the limits requires some $Q \in K$ with $QH_k(X_\infty)Q^{-1} = H_k(Y_\infty)$ for all $k \in I_u$. Consider $\Phi_K: K \times G^{r_u} \rightarrow G^{r_u} \times G^{r_u}$. Its rank at (Q, h) is $\dim K + r_u \cdot \dim G - s_K(h)$, where $s_K(h)$ is the dimension of the centralizer of h inside K .

We need this to be below the target dimension $2r_u \cdot \dim G$, that is: $r_u \cdot \dim G - \dim K + s_K(h) \geq 1$

- If K is a proper subgroup of G , then $\dim K < \dim G$ (a closed subgroup of the same dimension would contain the identity component, which is all of G since G is connected). Then $r_u \cdot \dim G - \dim K \geq \dim G - \dim K \geq 1$
- If $K = G$ (the shared holonomies are central, for example trivial), the situation is the LNC Impossibility Theorem with r_u in place of r : for $r_u \geq 2$ the margin is at least $(r_u - 1) \cdot \dim G \geq 1$, and for $r_u = 1$ the margin is $s_K(h) \geq \text{rank}(G) \geq 1$ by the maximal-torus argument.

In every case the image has measure zero by Sard's theorem (which states that the set of critical values of a smooth map between differentiable manifolds has a Lebesgue measure of zero), conditionally on h_s , hence also unconditionally. Since the minimum defining d is attained, $\epsilon_0 > 0$ almost surely. ■

When the floor can vanish. If I_u is empty (both agents share every transition and have identical substrates), $\epsilon_0 = 0$ is possible. This is a measure-zero coincidence such as digital twins. A substrate difference on a shared edge makes the cycles through that edge belong to I_u .

6. So what does it tell us

What can we reply to sceptics who still argue that all this does not solve the hard problem and that's not the hard problem they were asking about? There are two aspects of the explanatory gap. One is explaining the difference between 1st and 3rd person views (why does the 1st person view feeling like something at all), and the other one is finding an exact translation between two first-person perspectives.

The second question is the one that matters pragmatically in our lives. If we cannot understand subjective 1st-person experience in an objective way by measuring it, can we at least try to understand it by transforming a 1-person experience from one person to another? The answer to this is - as the theorems show - that exact equivalence of subjective experience is unattainable, and shared experience gets you only to a floor that is almost surely above zero. But even this minimal understanding gives us access to the larger question. As members of the same biological species, we know what it is like to be a human being that experiences pleasure and pain and all the other basic emotions. Each of us knows what it is like to feel good or bad, happy or sad. The longer we see what the other person went through, the more we understand what it must be like to be that person.

The answer to the first question (what it is like to feel something at all and why structure is accompanied by experience in the first place) can therefore be answered by everybody personally. This answer is in my opinion ultimately anchored in our biological nature. Our minds are inextricably linked to a single, unique, biological body. Our genes have created biomolecular control mechanisms - emotions - to control us. They tell us in every moment if something is good or bad, if something should be approached or not, if something should be consumed or not. These are in my opinion the irreducible dimensions of subjective experience.

They are linked to the needs of our biological body - mainly survival and reproduction. Whenever something is happening in nature in the biological world it is either supper or pairing time. The most basic emotions hunger and disgust (consume it-expel it), pleasure and pain (do it-stop it), love and fear (approach it-get away from it) are direct diametrical opposites of primary subjective dimensions.

Dimension	Biological Meaning	Operational Meaning
Pleasure vs. Pain	Tissue Maintenance vs. Tissue Damage	"Engage with this / Yes, do this" vs. "Protect yourself / No, stop this"
Hunger vs. Disgust	Energy Intake vs. Toxicity Avoidance	"Incorporate into Self" vs. "Expel from Self"
Love/Attraction vs. Fear	Reproduction vs. Threat Evasion	"Approach / Merge" vs. "Flee / Run away"

Thus, if we try to find out what it is like to feel something we eventually will hear behind the whole bio-molecular machinery of our emotions the whisper of our primary biological directive that is encoded in our genes. They are the ultimate judges if something should feel good or bad. The message of pleasure which our genes whisper is "This is going well. Continue, engage, consolidate this. Do this. You are in alignment with your primary directive".

Pleasure and pain are the most fundamental irreducible dimensions of subjective experience. They feel like they do because we have the bodies we have, which are built for survival and experience emotional valuations to guide our behavior. If we mean by the 'Hard Problem' explaining why a biological body built for survival experiences valence (pleasure/pain) as a directional force, then the answer is embodied homeostatic necessity — the irreducible dimensions of qualia are the physical orientation vectors of self-preservation.